

CURRICULUM VITAE (MAY 2018)

DR. RER. NAT. LUTZ BECKS

ADDRESS: Schillerstrasse 14
24116 Kiel
Phone: +49 (0)151 58533983

INSTITUTIONAL ADDRESS: Max-Planck Institute for Evolutionary Biology
Community Dynamics Group
August-Thienemann-Str. 2, 24306 Plön Germany
Phone: +49 (0)4522 763230, Email: lbecks@evolbio.mpg.de
www.evolbio.mpg.de/comdyn

DATE OF BIRTH: 21. April 1978 PLACE OF BIRTH: Jülich
CITIZENSHIP: Germany

PROFESSIONAL EXPERIENCE

since 10. 2011 Leader of an independent junior research group (*Emmy Noether program and Heisenberg Program, DFG*) at the Max Planck Institute for Evolutionary Biology.

11. 2010 – 10.2011 Postdoctoral Fellow at University of Cologne (Germany) with Prof. Hartmut Arndt (*Volkswagen Foundation Scholarship*).

10.2009 – 10.2010 Postdoctoral Fellow at University of Toronto (Canada) with Prof. Aneil F. Agrawal (*Volkswagen Foundation Scholarship*).

08.2006 – 09.2008 Postdoctoral Fellow at Cornell University, Ithaca NY (USA) with Prof. Nelson G. Hairston Jr. and Prof. Stephen P. Ellner.

EDUCATION

03.2004 – 10.2006 Ph.D. in Zoology at the Zoological Institute, University of Cologne (Germany) with Prof. Hartmut Arndt. Thesis title: "Complex population dynamics in microbial systems" (*magna cum laude*).

02.2002 – 02.2004 Studies and graduation at the University of Cologne (Germany) with a Diploma in Biology. Thesis title: "Chaotic dynamics – a common phenomenon in a two-prey-one-predator system? - Theoretical and experimental analyses of bacteria – protozoan interactions in chemostats" (*sehr gut*).

08.2001 – 01.2002 Erasmus exchange student at the University of Lund (Sweden).

10.1998 – 07.2001 Studies at the University of Cologne (Germany) in Biology.

PUBLICATIONS

Frickel, J., P. Feulner, E. Karakoc & **L. Becks**. *Population size changes and selection drive patterns of parallel evolution in a host-virus system*. **Nature Communications**, 1706.

Huang, W., Traulsen, A., Werner, B., Hiltunen, T. & **L. Becks**. 2017. *Dynamical trade-offs arise from species coevolution and lower the species diversity*. **Nature Communications**, 2059.

Ehrlich, E., **L. Becks** & U. Gaedke, 2017. *The equalizing trade-off: A key to explain species coexistence and co-occurrence*. **Ecology**, 98, 3188-3198.

- Hiltunen, T., Kaitala, V., Laakso, J. & **L. Becks**, 2017. *Evolutionary contribution to coexistence of competitors in microbial food webs*. **Proceedings of the Royal Society B**, 284, 1864.
- Frickel, J., L. Theodosiou & **L. Becks**, 2017. *Rapid evolution of hosts begets species diversity at the cost of intraspecific diversity*. **PNAS**, 201701845.
- Cairns, J., Frickel, J., Jalasvuori, M., Hiltunen, T. & **L. Becks**, 2017. *Genomic evolution of bacterial populations under co-selection by antibiotics and phage*. **Molecular Ecology**, 26, 1848-1859.
- Koch, H. & **L. Becks**, 2017. *The cost of facultative sex in a prey adapting to predation*. **Journal of Evolutionary Biology**, 30, 210-220. (Recommended by Faculty 1000).
- Cairns, J., **Becks, L.**, Jalasvuori, M. & T. Hiltunen, 2016. *Sublethal streptomycin concentrations and lytic bacteriophage interactively promote resistance evolution*. **Phil. Trans. R. Soc. B.**, 372, 20160040.
- Frickel, J., Sieber, M. & **L. Becks**, 2016. *Eco-evolutionary dynamics in a coevolving host-virus system*. **Ecology Letters**, 19, 450-459.
- Haafke, J., Abou Chakra, M. & **L. Becks**, 2016. *Eco-evolutionary dynamics select indirectly for sex in predator populations*. **Evolution**, 70, 641-652.
- Koch, H., Jeschke, A. & **L. Becks**, 2015. *Use of ddPCR in experimental evolution studies*. **Methods in Ecology and Evolution**, 7, 340-351.
- Hiltunen, T. Ayan, G. & **L. Becks**, 2015. *Environmental fluctuations restrict eco-evolutionary dynamics in predator-prey system*. **Proceedings of the Royal Society B**, 282. doi:10.1098/rspb.2015.0013.
- Becks, L.** & Y. Salehi-Alawi, *Using microevolution to explain the macroevolutionary observations for the evolution of sex*. In: *Macroevolution: Explanation, Interpretation and Evidence*, Volume 2, 279-299 (Springer).
- Hiltunen, T. & **L. Becks**, 2014. *Consumer co-evolution as an important component of the eco-evolutionary feedback*. **Nature Communications** 5, 5226, doi:10.1038/ncomms6226.
- Koch, H., Frickel, J. Valiadi, M. & **L. Becks**, 2014. *Why rapid, adaptive evolution matters for community dynamics*. **Frontiers in Ecology and Evolution** 2:17. doi: 10.3389/fevo.2014.00017/.
- Becks, L.** & A.F. Agrawal, 2013. *Higher rates of sex evolve under K-selection*. **Journal of Evolutionary Biology** 26, 900-905.
- Becks, L.** & H. Arndt, 2013. *Different types of synchrony in chaotic and cyclic communities*. **Nature Communications** 4, doi:10.1038/ncomms2355.
- Becks, L.** & A.F. Agrawal, 2012. *The evolution of sex is favoured during adaptation to new environments*. **PLoS Biology** 10, 1-11. (Recommended by Faculty 1000).
- Becks, L.**, S.P. Ellner, Jones, L. E. & N.G. Hairston Jr., 2012. *The functional genomics of an eco-evolutionary feedback loop: linking gene expression, trait evolution, and community dynamics*. **Ecology Letters** 15, 492-501. (Recommended by Faculty 1000).

Ellner, S.P. & **L. Becks**, 2011. *Rapid prey evolution in response to two predators*. **Theoretical Ecology** 4,133-152.

Becks, L. & A.F. Agrawal, 2011. *The effect of sex on the mean and variance of fitness in facultatively sexual rotifers*. **Journal of Evolutionary Biology**, 24, 656-664.

Becks, L., S.P. Ellner, Jones, L. E. & N.G. Hairston Jr., 2010. *Reduction of adaptive genetic diversity radically alters eco-evolutionary community dynamics*. **Ecology Letters**, 13, 989-997. (Recommended by Faculty 1000).

Becks, L. & A.F. Agrawal, 2010. *Higher rates of sex evolve in spatially heterogeneous environments*. **Nature**, 468, 89-92. (Recommended by Faculty 1000).

Jones, L. E., **Becks, L.**, Ellner, S.P., Hairston Jr. N.G., Yoshida,T. & G. Fussmann, 2009. *Rapid contemporary evolution and clonal food web dynamics*. **Phil. Trans. R. Soc. B.**, 364,1579-1591.

Becks, L. & H. Arndt, 2008. *Transitions from stable equilibria to chaos, and back, in an experimental food web*. **Ecology**, 89, 3222-3226.

Becks, L., Hilker, F., Malchow. H., Jürgens, K. & H. Arndt, 2005. *Experimental demonstration of chaos in a microbial food web*. **Nature**, 435, 1226-1229. (Recommended by Faculty 1000).

GRANTS AND AWARDS

Total: ~4,150,000 EUR

"Host virus coevolution – demography versus selection in the face of multiple stressors" German Research Foundation (DFG) **334,497 EUR**

"Indirect response to external drivers through trait variation in predator-prey systems" German Research Foundation (DFG), 2018 - 2021 **382,550 EUR**

Heisenberg Professorship from the German Research Foundation (DFG) at the Christian Albrechts University Kiel, 2017-2020) **373,000 EUR**

"Eco-evolutionary dynamics in evolving communities" German Research Foundation (DFG), 2017 - 2019 **233,650 EUR**

"Persistence mechanisms in a tripartite protist — giant virus — virophage system" (Betty and Gordon Moore Foundation) Investigator: L. Becks & M. Fischer (MPI Heidelberg), Jan. 2017 – Dec. 2108 part LB: **200,000 USD**

"Experimental Evolution and Community Dynamics - EECD 2016" (DFG, International Scientific Events) 2016 **10,200 EUR**

"Host virus coevolution – demography versus selection" Lead-Agency program" with SNF (P. Feulner EAWAG) as part of the Priority program by the German Research Foundation (DFG) SPP1819: "Rapid Evolution", Jan. 2016 - Dec. 2018 part LB: **322,328 EUR**

"Evolution in sexual and asexual populations" (Max-Planck Society), 2015 **80,000 EUR**

"Impact of the trade-offs on eco-evolutionary feedbacks in predator-prey systems" as part of the Priority program (DFG) SPP1704: "Flexibility matters: Interplay between trait diversity and ecological dynamics using aquatic communities as model systems (DynaTrait) "; Principal

Investigator: L. Becks & U. Gaedke (University of Potsdam,) Oct. 2014 - Oct. 2017

part LB: **187,805 EUR**

"Coevolving communities in spatially structured and heterogeneous environments" (DAAD/Academy of Finland); Principal Investigator: L. Becks & T. Hiltunen, Jan. 2014 - Dez. 2016

part LB: **10,963 EUR**

"Eco-evolutionary dynamics in complex systems" Emmy Noether Grant (DFG), Oct. 2011 - Oct. 2015

1,239,649 EUR

"The effects of sex on the coevolution and dynamics of a predator-prey system" (Volkswagen Foundation Postdoctoral fellowship), Oct. 2008 - Oct. 2011

193,000 EUR

"Studying complex dynamics: Contemporary rapid evolution in complex ecological communities" (James S. McDonnell Foundation); Principal Investigators: S. P. Ellner, N. G. Hairston Jr., L. Jones, L. Becks (all Cornell University, Ithaca), G. Fussmann (McGill University, Montreal), 2007 - 2011

449,459 USD

Others:

- Shortlisted for W3 position at University Mainz (2nd position; 2015).
- Shortlisted for professorship IST Austria (2016).
- Shortlisted for lecture at Queen Mary University London (2016).
- Young scientist award from the German Limnological Society (DGL), 2005.
- Excellent student scholarship for graduate studies from the University of Cologne (Germany).

INVITED PRESENTATIONS

I have been invited to present my research at internationally established conferences, symposia and advanced schools (**selected for the last 3 years, see below**).

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| 2018 | Sex uncovered: the evolutionary biology of reproductive systems, CONFERENCES JACQUES MONOD (France) |
| 2018 | Workshop "Experimental approaches to test for coevolution, Paris (France). |
| 2017 | Ringberg Symposium - Giant Viruses (Germany); Visions in Science, Berlin (Germany); Workshop "Origin of Biodiversity - Host-pathogen co-evolution in the genomic era" Gothenburg (Sweden). |
| 2016 | EVENT, Ghent (Belgium); ETH Zürich (Switzerland); IST Austria (Austria), Workshop "Experimental Evolution: Theory and Current Practices", Paris (France); Workshop "The genomic basis of eco-evolutionary change", Monte Veritas (Switzerland). |
| 2015 | European Phycological Congress, London (UK), German Ecological Society, Göttingen (Germany). |

SUPERVISION

Bachelor: Niklas Niehus (2015), Annika Friedrichsen (2016).

Master: Noemi Woltermann (2015), Sophia Wagner (2015), Aline Jeschkes (2016), Chris Gundlach (2016), Lasse Harder (2017), Ruben Hermann (ongoing).

PhD: Jens Frickel (defended 26.4.2016), Hanna Koch (defended 9.2016), Gökce Ayan (defended 11. 2017), Loukas Theodosiou (ongoing), Noemi Woltermann (currently on maternal leave). Elena Horas (ongoing).

Postdoc: Martha Valiadi (2.2013 -10.2013), Amrei Binzer (9.2013 - 9.2014), Heike Gruber *Aging and epigenetic modifications* (4.2014 - 1.2016), Vienna Kowallik (1.1.2016 - 31.1.2017), Ana del Arco: (1.4.2017-31.12.2018)

SCIENTIFIC SERVICES

- Faculty of the International Max Planck Research School Evolutionary Biology and member of several thesis committees
- Associate Editor Ecology Letters
- Associate Editor BMC Evolutionary Biology
- Associate Editor Frontiers in Ecology and Evolution
- Organization of the Workshop 'Bridging Theory and Experiments ~Evolving Communities' MPI Plön (2016)
- Organization of Mini-Symposium 'Experimental Evolution and Community Dynamics' University of Helsinki together with Dr. Teppo Hiltunen (2014, 2015) MPI Plön (2016)
- Organization of Symposium "Rapid evolution and population genetics" together with Dr. Teppo Hiltunen at the ESEB meeting in Lisboa (2013)
- Organization of Symposium "Ecology and Evolution of Sex" together with Hanna Koch at the ESEB meeting in Lausanne (2015)
- Organization of Graduate meeting of the German Zoological Society, MPI Plön (2017)
- Co-speaker of DFG priority program DynaTrait
- Co-organization of Kick-off meeting program DynaTrait (2014), annual Meeting (2015, 2016), international meeting (2017)
- Co-organization of Seminar series at the MPI Plön: Meselson Seminar on the Evolution of Sex
- Review for scientific journals: American Naturalist, BMC Biology, BMC Evolutionary Biology, Current Biology, Ecology, Ecology Letters, Evolution, Hydrobiologia, ISME Journal, Journal of Animal Ecology, Journal of Evolutionary Biology, Journal of Phycology, Limnology, Limnology and Oceanography, Molecular Ecology, Molecular Systems Biology, Nature Communications, PNAS, Trends in Ecology and Evolution
- Review or funding agencies: German Research Foundation (DFG), Swiss National Science Foundation (SNSF), Biotechnology and Biological Sciences Research Council (BBSRC), University of Bordeaux, University of Helsinki, FWO (Belgium), ERC.
- Outreach:
 - Darwin Day University of Kiel; Talk for 1200 high school students.
 - Public lectures in 2014, 2015 at the Max Planck Institute for Evolutionary Biology on the evolution of sex.
 - TV interview with public TV station (NDR) explaining the problem of the evolution of sex (2015).
- Scientific representative of the MPI Evolutionary Biology in the Max Planck Society.